V , E

Question 1: ..... 5 points

(a) **(2 points)** Complete the following table. MST = Minimum Spanning Tree.

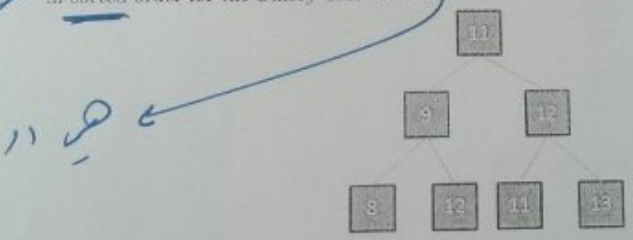
Algorithm	Worst Case Complexity
0-1 Knapsack Problem using Dynamic Programming	O(n*W)
0-1 Knapsack Problem using Brute Force	$O(2^n)$
Breadth First Search	O(|V|+|E|)
Depth First Search	O(|V| + |E|)
Kruskal's algorithm for finding MST	O(|E| log |E|)
Prim's algorithm for finding MST	O(|E| log(|V|))
Shortest path by Dijkstra	O(|V| · |E|)
Shortest path by Bellman Ford	

O(VE log N) → BFJ

O(V log N + E)

→ Priority Queue

(b) **(3 points)** Use Pre-order tree walk, In-order tree walk and Post-order tree walk to print all keys in sorted order for the Binary Tree below



sorted !!

....

....

Question 2: ..... 10 points

Consider the following instance of the 0.1 knapsack problem

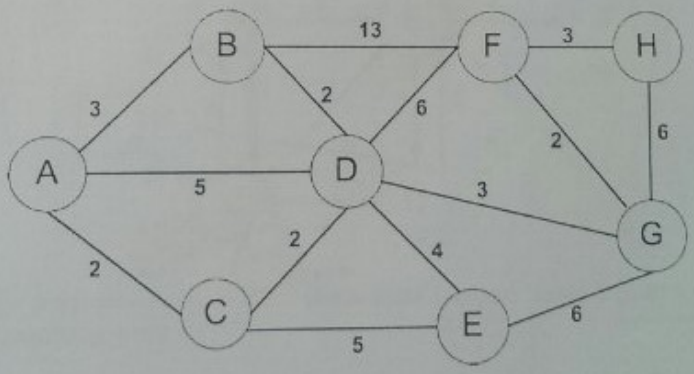
Item	1	2	3	4	5
Benefit	6	12	4	9	14
Weight	1	5	3	4	6

The maximum allowable total weight in the knapsack is $W = 10$.

Find an optimal solution for the above problem with the weights and benefits above using Dynamic Programming. Be sure to state both the value of the maximum benefit that you obtain as well as the item(s) that you need to obtain this benefit. Show all steps.

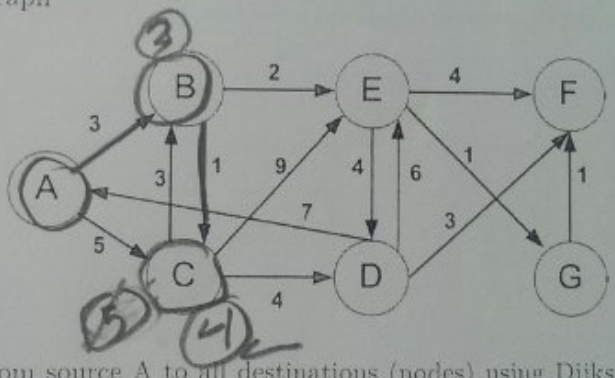
Question 3: ..... **8 points**

Find The Minimum Spanning Tree of the following graph using Kruskal's algorithm. Show the steps and the MST along with resulting length.



Question 4: ..... 8 points

Consider the following Graph



Find the shortest path from source A to all destinations (nodes) using Dijkstra's algorithm. Show all steps, tables and the resulting path.

Question 5:

9 points

(a) **(5 points)** Explain what are P, NP and NP-Complete problems. What is meant by "is P = NP"?

(b) Suppose that A and B, and that $A \leq_P B$ i.e. A can be reduced to B in polynomial time. Circle one answer for each statement below.

i. **(1 point)** If A ∈ NP-Complete and B is in NP then B ∈ NP-Complete
   A. True
   B. False

ii. **(1 point)** If B ∈ P, then A ∈ P
   A. True
   B. False

iii. **(1 point)** If A is NP-hard then B is NP-hard
   A. True
   B. False

iv. **(1 point)** If B is NP-hard then A is NP-hard
   A. True
   B. False

A → B

(b) Suppose that A and B , and that A ≤p B. Circle one answer for each statement below.

If A ∈ P then B ∈ P.

True False

If B ∈ P then A ∈ P.

True False

If A is NP-hard then B is NP-hard.

True False

If B is NP-hard then A is NP-hard.

True False

Question 5 (bonus Question): (6 points - 2)4)

(a) Let G = (V, E) be a weighted, directed graph with exactly one negative weight edge and no negative-weight cycles. Give an algorithm to find the shortest distance from s to all other vertices in V that has the same running time as Dijkstra.

Add positive value of negative weight to all weights then compute like Dijkstra

(b) A subset-sum problem is defined as follows:
A set S of positive integer and another positive integer t. Find a subset S' of integers in S such that their sum is t. For example, if S = {1,2,3,14,49,98} and t = 101, then the subset S' is {1,2,98}.

Assume that A subset-sum problem is NP-Complete. Show that the Knapsack problem is NP-Complete by reducing the subset-sum problem to Knapsack problem. [Hint: see that both problems are very similar].

Knapsack لإثبات أن subsetsum هي تحويل ①

∴ ② knapsack في np

∴ subsetsum في np-complete

∴ knapsack في np-complete

let S = { ~~1, 2, 3, 14, 49, 98~~ $w_1, w_2, w_3, \ldots w_n$ } ①

Knapsack as following

i	1	2		n
w_i	w_1	w_2		w_n
b_i	w_1	w_2		w_n

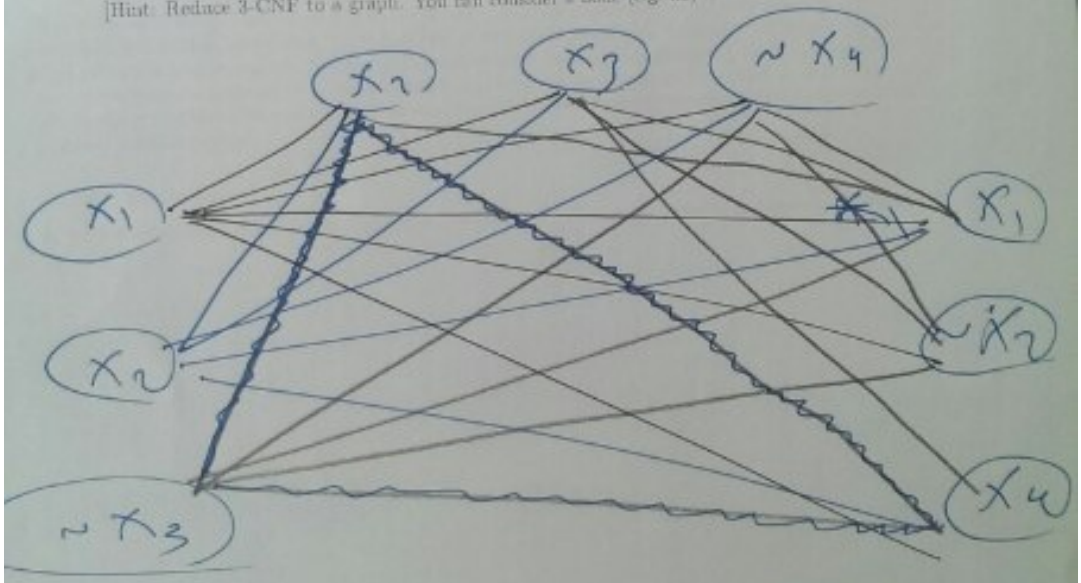
(b) (4 points (bonus)) An independent set in a Graph is defined as an independent set of vertices in a graph, no two of which are adjacent. That is, it is a set S of vertices such that for every two vertices in S, there is no edge connecting the two. The size of an independent set is the number of vertices it contains. Assume that independent set problem is in NP.

Assume the following 3-CNF [1] which is NP-Complete.

$(x1 \vee x2 \vee \sim x3) \wedge (x2 \vee x3 \vee \sim x4) \wedge (x1 \vee \sim x2 \vee x4)$ where $\sim$ means NOT function, $\vee$ means OR function and $\wedge$ means AND function.

Show that Independent set in a graph corresponding to the 3-CNF is NP-Complete. What is the size of the Independent set S (i.e. maximum number of nodes in S that are not adjacent)? Explain your answer.

[Hint: Reduce 3-CNF to a graph. You can consider a node (e.g. x2) is connected to (~x2) etc.]



$x_2 = 1$, $x_3 = 0$, $x_4 = 1$

[1] See Appendix for its definition